Homework 5 | Mirjabbar Badalov | MPA-SVT

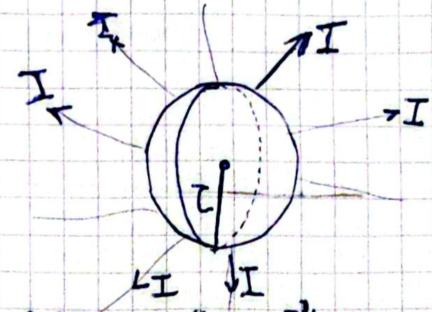
Problem 9:

A heated sphere $d = 1{,}9$ cm with luminous intensity $I = 95$ cd

Luminance $L = ?$

Luminosity $- ?$

Luminous flux $- ?$



$r = 0{,}95 \text{ cm} = 95 \times 10^{-4}$ m

$I = 95$ cd

The source radiates uniformly:

Then: $I = \frac{d\Phi}{d\Omega}$ $\quad \Omega$ will be 4π because of sphere

$$\left[\Phi = I \cdot 4\pi = 95 \cdot 4\pi = 1194 \text{ lm}\right]$$

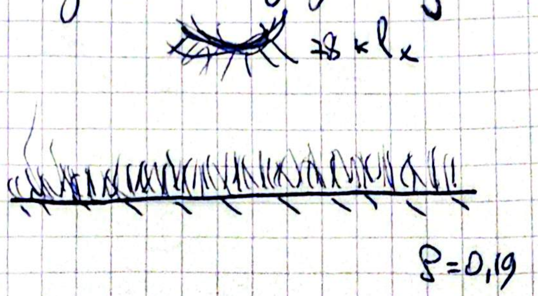
$$\left[L = \frac{I}{S}\right] \longrightarrow L = \frac{I}{\pi (d/2)^2} = \frac{4I}{\pi d^2} = \left[3.35 \times 10^5 \text{ cd/m}^2\right]$$

Luminosity $\rightarrow$ Luminous Exitance:

$$\left[H = \frac{\Phi}{S} = \frac{4\pi I}{\pi d^2} = \frac{4I}{d^2} = 1.05 \times 10^6 \text{ lm/m}^2\right]$$

Problem 10

Luminance of the grass patch under summer daylight 78 klx, with the reflectance of grass $\rho = 0,19$



$$L = \frac{E \cdot \rho}{\pi}$$

$$\boxed{L = \frac{78 \cdot 10^3 \cdot 0,19}{\pi} = 4,717 \ cd/m^2}$$